[CLASSIFIED . SUBMISSION ROUND]

THE CORPORATE HEIST.

Two hours. Ten thousand profiles. Five hundred names that matter.
Solution documentation for the Innov8 4.0 preliminary round.

FIELD	DETAIL
TEAM NAME	finalist
UNSTOP ID	finalist
MEMBERS	Harsh Pratap Singh, Rohit Kumar
DELIVERABLES	Ranked shortlist, Code pipeline, Solution documentation
OBJECTIVE	Identify Nightingale's top 500 applicants from the Vault using historical data + insider intel.

READ THIS FILE ALONGSIDE THE PIPELINE SOURCE CODE. STRICTLY CONFIDENTIAL.

FILE 01 // OBJECTIVE & ARCHITECTURE

01 EXECUTIVE SUMMARY

We engineered a 6-phase machine learning pipeline to identify Nightingale's top 500 candidates from the Vault. The architecture relies on a 6-model ensemble of gradient-boosted trees (LightGBM, XGBoost, CatBoost) trained on the Archive's historical performance data. This base score is then algorithmically adjusted to account for the current cycle's regime changes using the provided insider intelligence.

We deployed aggressive zero-tolerance filters to eliminate fabricated profiles (232 removed), duplicate candidate entries (244 removed), and individuals with notice periods exceeding two months (208 removed). Our post-model adjusters forcefully boost sustained open-source contributors and non-pedigree applicants with high technical scores, while severely penalizing title inflation.

VALIDATION METRICS

Dev set validation achieved a robust 62/150 overlap (41.3% precision@150). Total end-to-end pipeline runtime: 33 seconds (well within the 5-minute strict ceiling).

FILE 02 // DATA INTEGRITY

02 CLEANING & PARSING

The dataset was highly unstructured with over 20 fields requiring normalisation. Key standardisations include:

FEATURE	TRANSFORMATION PROTOCOL
Experience	Regex parsing of 15+ formats ("17.6 years", ">20", "Fresher", "69 months") into float years.
Compensation	Currency standardisation to Lakhs ("58.3 LPA" -> 58.3, "7.53 Cr" -> 753.0).
Notice Period	Mapped 14 text variants to discrete days (e.g., "Immediate" -> 0, "1 month" -> 30).
Performance	Unified numeric ("3/5") and text ("Outstanding") ratings onto a strict 1-5 scale.
KPI Met	Normalized 10+ boolean variants ("Y", "TRUE", "1") into strict binary 1/0 flags.
Education	Canonicalized degrees and fuzzy-matched institutes to flag IIT/NIT/IIIT status.

FILE 03 // ALPHA SIGNALS

03 WHAT DRIVES HIRES

Feature importance extraction from the ensemble, combined with dev set distribution analysis, revealed clear success vectors:

SIGNAL VECTOR	WINNERS VS OTHERS	IMPACT MULTIPLIER
Technical Assessment	66.9 vs 51.7	1.30x (Strongest Predictor)
KPI Achievement	63% vs 34%	1.86x Separation
Performance Composite	Derived Feature	Multiplicative combo: tech x apt x kpi
Aptitude Score	6.83 vs 5.72	1.19x Differential
Current CTC (Lower)	23.7 vs 35.3 LPA	0.67x (Cost-effective preference)
Skill Diversity	6.35 vs 5.04	1.26x Technical breadth proxy

Role geometry is non-uniform: Roles like Data Scientist (9.9% win rate), DevOps (8.0%), and ML Engineer (7.8%) dominate the top decile compared to Product Analyst (2.8%) and QA (1.9%).

FILE 04 // REGIME CHANGE

04 THE CURRENT CYCLE

The Vault regime fundamentally breaks from historical Archive patterns. We hardcoded the insider intelligence as post-model multipliers to shift predictions into the current regime reality:

INSIDER INTELLIGENCE DIRECTIVE	ALGORITHMIC ENCODING
Code Contributions Logging	>=30 merits +35% boost; >=15 merits +15% boost.
Title Inflation Backlash	VP/Director with <5yr experience -> 70-85% severe penalty.
Urgent Headcount Limits	Hard exclusion for notice periods > 60 days.
Non-pedigree Superstars	Non-traditional institute + tech>=80 -> +20% massive boost.
Historical Bias Removal	Metro/big-brand/pedigree artificially de-weighted.
Old-boys' Network Survival	Top 5 classic IITs retain a discrete +5% edge.

FILE 05 // SYSTEM PURGE

05 ZERO-TOLERANCE EXCLUSIONS

We developed strict exclusionary logic to enforce Nightingale's absolute rules. Any candidate failing these checks was permanently removed from the eligible pool.

FABRICATED PROFILES (232 PURGED)

A multi-dimensional fabrication score tracked timeline impossibilities (experience exceeding age - 18), extreme title inflation, career duration mismatches, and suspicious >5x CTC jumps. Threshold breached = deletion.

DUPLICATE ENTITIES (244 PURGED)

Detected via normalized email and phone matching via Union-Find. Collisions were resolved by keeping the profile with the highest technical assessment.

AVAILABILITY RULES (208 PURGED)

All notice periods exceeding 60 days were dropped per the insider mandate.

> **POOL STATUS**
> *668 total profiles excluded. 9,332 eligible candidates remained for final ranking.*

FILE 06 // TOPOLOGY

06 ENSEMBLE ARCHITECTURE

The core engine utilizes 60 engineered features across 9 categories. The ensemble comprises 6 gradient-boosted models, heavily weighted towards classification to directly optimize for top-5% discrete boundaries.

ALGORITHM	CONFIGURATION PARAMETERS	WT
LightGBM Regressor	800 rounds, lr=0.03, 127 leaves, L1/L2 reg	10%
XGBoost Regressor	800 rounds, lr=0.03, depth=7, regularized	10%
CatBoost Regressor	800 iter, lr=0.03, depth=7, l2=3.0	10%
LightGBM Classifier	800 rounds, binary AUC, unbalanced	25%
XGBoost Classifier	800 rounds, scale_pos_weight balanced	25%
CatBoost Classifier	800 iter, auto_class_weights=Balanced	20%

All predictions undergo rank-normalization (via scipy.rankdata) prior to weighted aggregation to prevent outlier distortion.

FILE 07 // DEAD ENDS

07 FAILED APPROACHES

- Sentence-transformers on recruiter_notes: NLP latency exceeded the 5-minute limit with minimal signal gain.
- Pure regression topology: Consistently failed to cleanly separate the top-5% threshold.
- Min-max scaling: Too susceptible to outlier skew compared to rank-based normalization.

FILE 08 // TECH STACK

08 RESOURCES & TOOLS

DEPENDENCY	UTILIZATION CONTEXT
LightGBM / XGBoost / CatBoost	Core gradient boosted ensemble architecture
pandas / numpy / scikit-learn	Vectorized data manipulation and preprocessing
rapidfuzz	Levenshtein distance matching for categorical canonicalization
Claude (Anthropic)	Rapid prototyping and strategic architecture

FILE 09 // PROTOCOL

09 HOW TO RUN

Execute the pipeline from the project root containing the four CSV files:

```
$ python code/main.py
```

PARAMETER	SPECIFICATION
Entry Point	code/main.py
Seed	42 (Deterministic across all libraries)
Compute Cost	~33 seconds (CPU only, 2 cores)
Output Target	submission.csv (Exactly 500 ranked rows)